FEDERAL SPECIFICATION

GLUEPOTS, ELECTRIC (PORTABLE, DRY AND WET)

This specification was approved on the above date by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers a portable electrical appliance in which animal glue is heated to a preset temperature and there automatically maintained by means of a thermostat and electrical resistance coils.

1.2 Types.—Pots shall be of the following types with automatic temperature control:

Type I. —Dry type
Type II.—Wet type (water jacket)

1.3 Sizes.—Pots shall be of the following sizes:

Size 1.—One-quart capacity
Size 2.—Two-quart capacity
Size 3.—Four-quart capacity

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

2.1 Specifications.—The following Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

C–G–451—Glue; Animal (for) Woodworking. (See 4.5.2.3.)
QQ–W–486—Wire and Ribbon; Resistance. (See 3.8.)

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

2.2 Other publication. — The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Standard (No. 499) for Electric Heating Appliances.

(Copies of Underwriters' Laboratories, Inc., standards may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Ill.; or 500 Sansome Street, San Francisco 11, Calif.)

3. REQUIREMENTS

3.1 Underwriters' Laboratories, Inc. — Gluepots, electric, delivered under this specification shall comply with the applicable requirements of Underwriters' Laboratories, Inc., Standard for Electric Heating Appliances (No. 499). The bidder shall submit proof to the purchasing agency that the gluepots, electric, he proposes to furnish under this specification conform to the standards of the Underwriters' Laboratories, Inc., regarding fire and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc., will be accepted as conforming to this requirement.

3.1.1 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the purchasing agency that the gluepots, electric, conform to the published

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standards, including methods of test, of the Underwriters' Laboratories.

3.1.2 Compliance with the above preliminary requirements, regarding fire and casualty hazards, does not absolve the bidder from complete compliance with the other requirements of this specification in order to secure the acceptance of his gluepots, electric.

3.2 **General description.** — Each gluepot shall consist of a readily removable inner pot and portable electric heater housing with flexible supply cable.

3.3 **Voltage rating and frequency.**—Pots shall operate on either 115 or 230 volts and on either direct or alternating 60-cycle current, as specified in the invitation for bids.

3.4 **Heater housing.**

3.4.1 *General description.* — The heater housing shall be compact and substantial in construction. Outer surfaces, if of iron or steel, shall be galvanized or japanned.

3.4.2 *Handle.*—Each pot shall be provided with a bail-type handle.

3.4.3 *Type I, dry type.*—The housing shall be of double-wall construction. The inner wall shall be of a suitable seamless durable metal and shall be plated to resist corrosion. Suitable insulation shall be used between the outer and inner walls.

3.4.4 *Type II, wet type.*—The housing may be of single- or double-wall construction. The surface in contact with the water must be impervious to water and must not be adversely affected by potable water. Suitable insulation shall be used in the walls of double construction.

3.5 **Inner pot.**—The inner pot shall be of seamless construction and shall be of copper, copper-nickel alloy, aluminum, aluminum alloy or corrosion-resisting steel, and shall have a smooth polished interior. It shall be provided with a suitably secured attached bail handle, brush wiper, and top flange which shall extend beyond the outer edge of the outer pot. All material shall be of such quality and thickness as to be rigid and durable. Unless the gluepot is of such design as to prevent glue from seeping between inner pot and inner side of heater, a reinforcement or other suitable means shall be fitted on the inner pot in order to permit its forcible removal from the housing in case the pot sticks.

3.6 **Lid.**—A steel lid shall be furnished. It shall be not less than 0.015 inch thick, shall be provided with a handle or knob, and shall be suitably plated to resist corrosion. Such lid shall be either portable or hinged and, when pot is not in use, shall exclude dust and dirt.

3.7 **Name plate.** — Each gluepot shall be stamped or fitted with a suitable name plate giving the manufacturer's name, the manufacturer's serial number, the type, the size and the capacity in quarts, the rated volts and amperes (maximum) or volts and watts (maximum), and specification number.

3.8 **Heating element.**—Each pot shall be provided with a properly arranged electric heating element, so located and insulated thermally as to prevent fire risk or damage to a wood support-surface when operating at the maximum temperature conditions at rated voltage. Heating elements shall be made of type D or E resistance material in accordance with the requirements of QQ-W-486.

3.9 **Thermostat.**

3.9.1 *Use.*—The temperature of the glue shall be automatically maintained by one or more suitable thermostats within the working temperature limits under the conditions prescribed in 4.5.1.

3.9.2 *Temperature setting.*—The tempera-

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ture setting of the thermostat shall be the mid-point between those temperatures related to the making and breaking temperatures.

3.9.3 *Working temperature limits.*—Working temperature limits as controlled by the thermostats shall be from 140° to 160° F.

3.9.4 *Sensitivity.*—Sensitivity of the thermostat shall be plus or minus 3° F. for the temperature setting over the range of 140° F. through 160° F. (See 4.5.1.)

3.9.5 *Design.* — Each thermostat shall be quick break, and shall properly make and break contacts in the heater supply circuit at low and high predetermined temperatures of the inner pot contents. All contacts on thermostats shall be provided with suitable hardened alloy or silver tips and arc absorbers. The mechanism on the thermostat for temperature-setting adjustment shall be both easily accessible and adjustable throughout the temperature range of 140° F. to 160° F.

3.10 Electrical fittings. — Pots shall be furnished with a separable electrical connection at the pot. This fitting shall be shielded or armored against drip and shall be attached permanently to the pot. A fitting of the recess type flush with the outer jacket will be acceptable.

3.11 Electrical cable. — Pots shall be furnished with 8 feet of rubber-jacketed heater cord having a one-piece soft-rubber attachment plug cap with finger grip at one end and composition cord connector body at other end. Unless the attachment plug is moulded on, a cord grip shall be provided on the plug.

3.12 Insulation, heat and electrical. — Heat and electrical insulation in proximity to parts subject to temperatures in excess of 49° C. (120° F.) shall be of flameproof, slow-burning, heat-resisting material.

3.13 Dielectric strength.—Dielectric material shall meet the requirements of 4.5.5.

3.14 Electrical insulation resistance.—The electrical insulation shall meet the resistance tests of 4.5.6.

3.15 Performance.

3.15.1 *Warming-up-time.* — The heater, when operated at rated voltage, shall distribute sufficient heat to the inner pot to melt the contents so that, when stirred, a uniform viscosity may be obtained in the time as given in 4.5.2.4.

3.15.2 *Temperature differential, quarter-full pot.*—The heater, when operated at rated voltage, shall meet the requirements of 4.5.3.

3.15.3 *Energy consumption.*—The heater, when operated at rated voltage, shall meet the requirements of 4.5.4.

3.16 Workmanship. — Workmanship shall be of first class.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection lot.—All pots of the same size, class, and type presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.2 Sampling for lot acceptance.

4.2.1 *Sampling for inspection.*—A random sample of pots shall be taken from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements specified in table I (in accordance with MIL-STD-105).

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TABLE I.—*Sampling for visual and dimensional inspection, AQL (approx.) = 1.5 percent defective*

Number of pots in inspection lot	Number of pots in sample	Acceptance number	Rejection number
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1,300	75	3	4
1,301 and over	110	4	5

4.3 Inspection.

4.3.1 *Visual and dimensional inspection.*—Each of the sample pots selected in accordance with table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with the requirements of this specification. Any pot in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective pots in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming pots. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with the specification requirements.

4.5 Test procedures.

4.5.1 *Temperature setting and sensitivity test.*

4.5.1.1 *Test conditions.*—Initial temperature as per 4.5.2.1; water level as per 4.5.2.2, for type II; glue level as per 4.5.2.3; and standard thermometer (mercury-in-glass), Bureau of Standards approved.

4.5.1.2 *Method of test.*—Meet conditions of 4.5.1.1. Heat pot normally for 3 hours. During the last 15 minutes immerse the thermometer approximately 3 inches and stir the glue constantly. Take temperature readings at intervals of 1 minute and plot results on graph paper, time being on the abscissa and temperature being on the ordinate. The curve must simulate a sine wave. The midpoint temperature between any peak and the adjacent valley temperature will be the temperature setting of the thermostat. The sensitivity of the thermostat shall be the total difference between the peak and valley temperatures. Adjust the temperature-setting mechanism to check the sensitivity at 140°, 150° and 160° F. In no instance shall the sensitivity be more than 6° F.

4.5.2 *Warming-up-time test.*

4.5.2.1 *Initial temperature.*—All parts and contents shall be at room temperature, i.e., 68° to 77° F.

4.5.2.2 *Water level.*—In case of type II, the water level in the jacket shall be at least up to the level of the glue or at least ½ inch from the top of the water compartment, whichever is the highest.

4.5.2.3 *Glue level.*—The pot shall be filled to ¼ inch from the brim with an 11-percent solution of grade J2 glue, conforming to Federal Specification C–G–451, which shall have been presoaked 16 hours at room temperature.

4.5.2.4 *Time limits.*—While being stirred as necessary, the starting time or the warming-up period of the 1-quart and 2-quart sizes shall not exceed 45 minutes to attain 140° F. glue temperature with the portable cover not in place. The 4-quart size shall require not more than 65 minutes for the same warming-up period with the portable lid not in place.

4.5.3 *Temperature-differential test, quarter-full pot.*

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4.5.3.1 Repeat test of 4.5.1 except pot shall be one-quarter filled with glue.

4.5.4 *Energy-consumption test.*

4.5.4.1 *Initial conditions.* — Same as for 4.5.1.

4.5.4.2 The electric energy shall be measured by a watt-hour meter of an accuracy of less than 10 percent error. The energy expenditure during 7 hours' operation, starting cold, shall not exceed that specified in table II for a 160° F. temperature setting.

TABLE II.—*Energy consumption*

Size	Total energy expenditure	
	Type I	Type II
Quarts	*Kilowatt hours*	*Kilowatt hours*
1	0.55	0.65
2	0.80	1.10
4	1.50	2.00

4.5.5 *Dielectric strength test.*—Each pot, while hot, shall be subjected to a dielectric test, using a 60-cycle alternating voltage of 1,000 volts, impressed between the current-carrying parts and the housing and pot, having practically a sine-wave form applied for 1 minute without rupturing the dielectric.

4.5.6 *Electrical insulation resistance test.*— Each pot, while hot, shall be tested with a 1,000 volt megger, or equivalent instrument, where the voltage is impressed in the same manner as 4.5.5, and such electrical insulation resistance shall be at least one megohm.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, substantial packaging in accordance with standard commercial practice is acceptable under this specification.

5.2 Packing, domestic.—Unless otherwise specified, the subject commodity shall be delivered with such packing as shall conform to the requirements of freight classification rules of lawfully operated common carriers in effect at the time of shipment, at the lowest rate to the point of delivery, provided such packing protects against the usual hazards of transportation of lawfully operated common carriers and shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.2.1 If crates are used, the pots shall be provided with protection against damage from dust and moisture.

5.3 Marking.

5.3.1 *Packages.*—Packages shall be marked with the name of the article, the type and size, and the quantity contained therein, as defined by the contract or order under which shipment is made, the voltage and current rating, the name of the contractor and the number of the contract or order.

5.3.2 *Shipping containers.*—Shipping containers shall be marked with the name of the article, the type, the size and the quantity contained therein, as defined by the contract or order under which shipment is made, the voltage and current of pots, the name of the contractor, the number of the contract or order and the gross weight.

6. NOTES

6.1 Ordering data. — Contracts, orders, requisitions and invitation for bids, should specify the following:

Type (see 1.2)
Size (see 1.3)
Voltage and frequency (see 3.3)

6.1.1 Purchasers should also exercise any options offered herein. (See 5.1, 5.2, 7.2.2.2.2, and 7.2.3.1.1.)

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related

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Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following departmental specifications, of the issues in effect on date of invitation for bids, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and other publications.*

Federal Specifications:

NN-B-591—Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).[1]

NN-B-601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.[1]

NN-B-621—Boxes; Wood, Nailed and Lock-Corner.[1]

NN-B-631—Boxes; Wood, Wirebound (for Domestic Shipment).[1]

UU-C-843—Cushioning Materials, Cellulosic.[1]

LLL-B-631—Boxes; Fiber, Corrugated (for Domestic Shipment).[1]

LLL-B-636—Boxes, Fiber, Solid (for Domestic Shipment).[1]

Military Specifications:

JAN-P-105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.[2]

JAN-P-106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.[2]

JAN-B-107—Boxes, Wood, Wirebound (Overseas Type).[1]

JAN-P-108—Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.[1]

JAN-P-116—Packaging and Packing for Overseas Shipment — Preservation, Methods of.[3]

JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.[1]

JAN-P-138—Packaging and Packing for Overseas Shipment — Boxes, Wood, Fiberboard-Lined.[1]

JAN-P-140—Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case-Liner.[1]

U. S. Army Specifications:

94-40645—Marking; Exterior, Domestic and Export Shipment, by Contractors.[4]

100-2—Standard Specification for Marking Shipments by Contractors.[3]

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.[1]

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Army.*

7.2.2.1 *Preparation for overseas shipment.*

[1] Applicable to Navy purchases only.
[2] Applicable to Army and Navy purchases only.
[3] Applicable to Army purchases only.
[4] Applicable to Air Force purchases only.

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7.2.2.1.1 *Preservation and unit packaging.*—Gluepots shall be individually preserved and unit-packaged in conformance with Military Specification JAN–P–116 and appendices thereto, method IA–2, without preservative, using snug-fitting fiberboard boxes, compliance symbol as applicable for the weight of the contents, as the containers. The dipcoat-sealed packages shall be overwrapped with kraft paper suitably secured.

7.2.2.1.2 *Packing.* — Gluepots of like description shall be packed together in quantities not to exceed 200 pounds gross weight in snug-fitting nailed wood boxes conforming to Military Specification JAN–P–106, style 2, 2½, or 3, or wood-cleated plywood boxes conforming to Military Specification JAN–P–105, style A or B, for type 2 load. Suitable cushioning and blocking shall be provided to prevent movement within the boxes. The boxes shall be closed and strapped in conformance with the appendix to the applicable box specification.

7.2.2.2 *Marking.*

7.2.2.2.1 *For identification.*—Each package, box, and overwrap shall be marked or labeled to indicate the item and quantity contained therein.

7.2.2.2.2 *For shipment.*—In addition to any special marking required by the contract or order, shipping containers for the Army shall be marked in accordance with U. S. Army Specification 100–2.

7.2.3 *Air Force.*

7.2.3.1 *Marking.*

7.2.3.1.1 *Exterior shipping container.*—In addition to any special marking required by the contract or order, each exterior shipping container shall be marked in accordance with U. S. Army Specification 94–40645.

7.2.3.1.2 *Interior package.*—Each interior package shall be legibly marked with the following information:

Gluepot, Electric Portable
Type
Size
Volts
Current
Fed. Spec. W–G–441
Manufacturer's part No.
Manufacturer's serial No.
Order No.
Stock No.
Manufacturer's name or trade-mark

7.2.4 *Navy.*

7.2.4.1 *Inspection procedures.*—For Naval purchases, the general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.4.2 *Packaging.*—The inner pot shall be cushioned with cellulosic wadding conforming to Federal Specification UU–C–843 and nested in the outer pot. One complete unit shall be wrapped with flexible, waterproof barrier-material conforming to Military Specification JAN–P–125. The seams and closures shall be sealed with water-resistant adhesive conforming to Military Specification JAN–P–140, and shall have a water-resistance equal to that of the body material, and the unit packages shall be placed in a corrugated or solid fiberboard box conforming to Federal Specification LLL–B–631 or LLL–B–636, respectively.

7.2.4.3 *Packing.*

7.2.4.3.1 *For domestic shipment.* — Portable electric gluepots, packaged as described in 7.2.4.2, shall be packed in wood-cleated-fiberboard, wood - cleated - plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber boxes conforming to Federal Specification NN–B–591, NN–B–601, NN–

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B-621, NN-B-631, LLL-B-631, or LLL-B-636, respectively. The gross weight of fiberboard boxes shall not exceed approximately 90 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

7.2.4.3.2 *For overseas shipment.*—Portable electric gluepots, packaged as described in 7.2.4.2, shall be packed in cleated-plywood, nailed-wood, wirebound, fiberboard, or fiberboard-lined wood boxes, conforming to Military Specification JAN-P-105, JAN-P-106, JAN-B-107, JAN-P-108, or JAN-P-138, respectively. The gross weight of wood boxes shall not exceed approximately 90 pounds, and of fiberboard boxes, approximately 70 pounds.

7.2.4.3.2.1 Contents shall be enclosed within a sealed waterproof bag, or its equivalent, made from material conforming to Military Specification JAN-P-125, for case-liner material. Seams and closures shall be sealed with adhesive conforming to Military Specification JAN-P-140.

7.2.4.4 *Marking.*

7.2.4.4.1 *Interior packages.*—Each interior package shall be durably and legibly marked with the following information:

Material description
Supplier or manufacturer
Contract or order No.
Fed. Spec. W-G-441
Quantity contained
Date packed
Service stock No.

7.2.4.4.2 *Shipping containers.*—In addition to any special marking required by the contract or order, marking of shipping containers shall be in accordance with Navy Shipment Marking Handbook.

7.2.4.5 *Ordering data.*—Procurement documents should specify whether domestic or overseas packing and marking are required. (See 7.2.4.3 and 7.2.4.4.)

NAVY INTEREST: A *MC* Sh S Y

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